Charles Nguyen

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Education

University of California, San Diego — *La Jolla, CA*
Bachelor of Science in Computer Science — *June 2026*

Technical Skills

Languages: TypeScript, JavaScript, Rust, Python, SQL
Frontend: React, React Native, Expo, Astro, TanStack Start
Backend & Infra: Convex, Firebase, Docker, Linux, Google APIs
Auth & AI: OIDC, OAuth 2.0, JWT, Logto, OpenRouter, Okta Certified Professional

Work Experience

ElitAxis LLC — *San Diego, CA (Remote)*
Software Engineer (Contract) — *Apr. 2026 – Present*

- Unified multi-server access through Warpgate; integrated Pocket ID via OpenID Connect (OIDC).
- Built Astro/Convex client applications with Stripe payments and custom shipping and fulfillment integrations.
- Co-developed a CMS for clients to manage website content, products, and business operations.

Halıcıoğlu Data Science Institute, UC San Diego — *Hybrid*
HDSI Lab 3.0 Fellow — *Nov. 2025 – June 2026*

- Built the lab's responsive Astro website with Markdown-driven project and event pages, dynamic member listings, and tables of contents; drafted the Privacy Policy to support COPPA compliance.
- Built an Astro/Convex platform for high-school partnerships, enabling account-free code sharing and debugging with version history, syntax highlighting, structured log visualization, and diff comparison.

IEEE Student Branch at UC San Diego — *San Diego, CA*
Webmaster — *Oct. 2023 – May 2026*

- Architected and maintained a centralized member and officer platform using TanStack Start, TypeScript, Convex, and Google APIs, supporting events, reimbursements, sponsorships, analytics, resume access, and user administration.
- Implemented organization-wide authentication using Logto, OpenID Connect, JWTs, and role-based access control, securing workflows and permissions for over 800 members and over 80 officers.

Technical Projects

Aulora | *TypeScript, React, React Native, Tauri, Convex* | aulora.spwnd.dev — *Sept. 2026 – Present*

Built self-hosted team chat with messaging and voice/video calls for web, desktop, iOS, and Android.

- Implemented shared client logic, server-side encryption, role-based access, and Docker deployment.

ChopChop | *TypeScript, React Native, Expo, Convex, OpenRouter* | chopchop.spwnd.dev — *Feb. 2026 – Present*

Built a receipt-scanning and expense-splitting app for iOS and Android; beta on TestFlight and Google Play.

- Implemented OpenRouter vision extraction with 98% accuracy and authenticated groups using Convex and Better Auth, reducing manual entry by 3–5 minutes per receipt.

IEEE Presence Tracker | *Rust, Convex, Raspberry Pi, BlueZ, Docker* — *Jan. 2026 – June 2026*

Built a Raspberry Pi Bluetooth presence system with Rust daemon, Convex backend, and web dashboard for app-free check-ins.

- Implemented l2ping and connect-probe detection with hysteresis/flap freeze, syncing attendance to Convex and Discord/Slack.

Leadership & Activities

IEEE Student Branch at UC San Diego — *San Diego, CA (Hybrid)*
Vice Chair of Projects (Student Leadership) — *Jan. 2026 – May 2026*

- Secured $10,000+ for TritonBots hardware and competition travel; coordinated scrimmages with Georgia Tech.
- Oversaw robotics, supercomputing, and design programs for hundreds of students, coordinating leads and milestones.
- Standardized project bills of materials, parts allocation, inventory tracking, and Open Access Hours logs.